



Karachi Institute of Economics and Technology
College of Computing and Information Sciences
FINAL YEAR PROJECT – Fall-2026
PROPOSAL

SEMESTER	FALL 2026	YEAR	2026-2027
Degree Program	BSCS/BSAI/BSSE/BSCYBSE		

TITLE OF PROPOSED PROJECT
TRIMLY

Project Category (choose one)	☒ Product based	☐ Research-based
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SUPERVISOR INFORMATION (To be filled by the office)			
Supervisor Name:		Organization/ Designation	
Contact No:		email:	

STUDENT(S) INFORMATION			
S#	Student ID	Name	
1	15865	Muhammad Zain	
Contact No:	03302492534	email:	mz0187914@gmail.com
Expertise	React Native, Node.js, Express.js, React.js, PostgreSQL, Tailwind CSS		
2	15925	Ali	
Contact No:	03132088955	email:	muhammadaliasghar905@gmail.com
Expertise	React Native, Node.js, Express.js, React.js, MongoDB, n8n, Python, Documentation		
3	15932	Hassan Ahmed Siddiqui	
Contact No:	03368894374	email:	hassansiddiqui.1959@gmail.com
Expertise	Next.js, React.js, Kotlin, Python, SQA		
4	15883	Muhammad Muzammil	
Contact No:	03172855256	email:	muzammil.muhammad7782@gmail.com
Expertise	Node.js, Express.js, React.js, MongoDB, PostgreSQL, Bootstrap, Documentation		

PROJECT AREA/TOOLS	
Tools Required:	Visual Studio Code, React Native, Node.js + Express.js, MongoDB Atlas, n8n, DeepInfra API, Figma
Area/Specialization:	Mobile Application Development, AI & Intelligent Systems, Business Process Automation

SUMMARY OF PROPOSED PROJECT (MAXIMUM 400 WORDS)
Getting a haircut in Pakistan should be easy. Walk into any salon on a Saturday afternoon, and the truth becomes clear. Plastic chairs are lined up. No idea how long the wait will be. No clue if the barber is here today. You wait. Either way you don't get what you want, which is simply a haircut. Trimly changes that. It is a platform that brings every salon, every barber, and every booking into one connected system. A customer opens the app, finds a salon nearby, sees prices, picks their barber, and books in under two minutes, all from their phone while sitting on their couch. Barbers get their own tools to manage their schedules. Salon owners get reports about busy times, money made, and weekly summaries without having to do any work themselves. What makes Trimly different is the Intelligence behind it. The platform learns from every booking. It suggests the barber. It recommends the time. It warns about no-shows before they happen. It sends reminders through WhatsApp when it's time for the visit. Trimly does not just digitize the salon experience, it transforms it. Owners spend less time managing and more time growing. Customers stop waiting and start planning. Pakistan's grooming industry, for the first time, runs on intelligence.

PROJECT OBJECTIVE(S)/OUTCOMES

- 1. Unified Salon Ecosystem:** To develop a multi-role mobile platform that connects customers, barbers, and salon owners within a unified salon ecosystem.
- 2. Real-Time Discovery and Booking:** To enable customers to discover salons, view transparent service pricing, and book appointments with their preferred barber in real time.
- 3. Barber Scheduling Interface:** To provide barbers with a dedicated scheduling interface for managing daily appointments and availability.
- 4. Business Analytics Dashboard:** To equip salon owners with a business dashboard offering revenue analytics, staff performance tracking, and booking trends.
- 5. AI Recommendation Engine:** To integrate an AI-powered recommendation engine that suggests barbers and services based on individual customer booking history.
- 6. No-Show Risk Scoring:** To implement a heuristic no-show risk scoring system that alerts barbers before high-risk appointments occur.
- 7. Automated Post-Visit Engagement:** To automate post-visit customer engagement through an n8n pipeline including review requests, loyalty point updates, and WhatsApp rebooking reminders.
- 8. Review Sentiment Analysis:** To incorporate sentiment analysis on customer reviews, generating automated alerts for salon owners when negative patterns are detected.
- 9. Secure Payment Integration:** To integrate a secure payment gateway supporting JazzCash and Easypaisa for deposit collection at the time of booking.
- 10. AI-Powered Hairstyle Recommendation:** To develop an AI-powered hairstyle recommendation feature that uses vision-language AI to analyze a customer's uploaded photo together with their chat-based style request, and generates personalized hairstyle suggestions accompanied by curated reference images.
- 11. Loyalty Points System:** To implement a loyalty points system that rewards repeat customers and encourages long-term retention within the platform.
- 12. Local Market Data Integration:** To collect and utilize real salon data from Karachi-based salons to ensure platform authenticity and relevance to the local market.

SIMILAR APPLICATIONS (SIMILARITY/DISSIMILARITY)				
S#	Name of App	Link (URL/Play Store)	Similar Features	New Features
1	Fresha	https://www.fresha.com/en-GB	• 24/7 online booking • Salon/spa-specific appointment calendar • Integrated card payments and no-show protection • Automated appointment reminders • Staff mobile notifications	• AI-powered barber and service recommendation engine • Heuristic no-show risk scoring • Sentiment analysis on reviews with owner alerts • AI-powered hairstyle recommendation via photo and chat • JazzCash/Easypaisa local payment support
2	Booksy	https://booksy.com/en-us/	• 24/7 online booking • Email/SMS auto-notifications • Staff scheduling and payroll • Client database and history • Secure in-app payments	• Karachi-based local salon data integration • AI recommendation and no-show risk scoring • WhatsApp-based automation pipeline • AI-powered hairstyle recommendation feature • JazzCash/Easypaisa local payment support
3	Vagaro	https://play.google.com/store/apps/details?id=com.vagaro	• Integrated payments and deposit collection • Drag-and-drop calendar • Multi-channel booking • Loyalty points and favourites • No-show/cancellation fees	• AI-driven personalization • WhatsApp Business API rebooking automation • AI-powered hairstyle recommendation feature • Localized Pakistani payment gateways and Karachi salon data

FUNCTIONAL FEATURES (Must be in Bullet Form)

For FYP 1 (At Least 70%):

- Design mockup and prototype (Figma)
- Database schema design (ERD)
- Single authentication system with role-based JWT routing (Customer / Barber / Salon Owner)
- Customer registration, login, and profile management
- Salon discovery with area-based filtering and Google Maps integration
- Service menu with transparent pricing per salon
- Barber selection with availability status
- Time slot booking system with appointment code generation
- Barber dashboard for daily schedule, availability toggle, and marking service as completed
- Salon owner dashboard for staff management and service price updates
- Ratings and reviews per barber
- Loyalty points system that rewards customers on every completed booking
- In-app chat between customer and salon
- Favorite option to save preferred barbers and salons
- Cancellation and reschedule policy engine with refund logic
- Push notifications for booking confirmation and appointment reminders
- AI-powered barber and service recommendation engine
- Smart slot suggestion based on historical booking density
- Heuristic no-show risk scoring on barber dashboard
- AI-based off-peak discount suggestions for salon owners
- Sentiment analysis on customer reviews with owner alert system
- n8n automation pipeline for post-visit review requests and loyalty points updates
- Revenue and booking trend charts for salon owner
- Real salon data collection and integration (Karachi-based)

For FYP 2 (Remaining 30%):

- JazzCash / Easypaisa sandbox payment integration
- WhatsApp Business API integration via 360dialog
- n8n automation pipeline for WhatsApp rebooking reminders
- AI-powered hairstyle recommendation via one-time photo upload and chat-based style requests, using vision-language AI
- Curated reference image suggestions for recommended hairstyles
- Weekly automated business report generation and email delivery to salon owner
- Smart rebooking reminder system via WhatsApp after service cycle completion
- Final testing, bug fixes, and performance optimization
- Complete deployment: Render (backend), MongoDB Atlas, Koyeb (n8n), Expo EAS (APK)
- Demo video (4-5 minutes)
- Poster / Panaflex preparation (2x5 ft)
- Complete FYP report (hard copy)

FYP Report at the time of FYP1 declaration (Soft Copy)

- ☐ Chapter 1 : Introduction
- ☐ Chapter 2 : Literature Review/Process Review
- ☐ Chapter 3 : Analysis Diagrams

For FYP 2:

FYP Report at the time of FYP2 declaration (Hard Copy)

- ☐ Complete FYP Report
- ☐ Panaflex (2x5 ft.)

Additional Features by Jury Members:

Sustainable Development Goals (SDG)

1. ☐ No Poverty
2. ☐ Zero Hunger
3. ☐ Good Health and Well-being
4. ☐ Quality Education
5. ☐ Gender Equality
6. ☐ Clean Water and Sanitation
7. ☐ Affordable and Clean Energy
8. ☒ **Decent Work and Economic Growth**
9. ☒ **Industry, Innovation, and Infrastructure**
10. ☐ Reduced Inequalities
11. ☒ **Sustainable Cities and Communities**
12. ☐ Responsible Consumption and Production
13. ☐ Climate Action
14. ☐ Life Below Water
15. ☐ Life on Land
16. ☐ Peace, Justice, and Strong Institutions
17. ☐ Partnerships for the Goals

For Proposal Defense Purpose

PROPOSED ADVISORY COMMITTEE		
S#	Faculty Name	Signature
1		
2		
3		
4		

FYP COMMITTEE			
S#	Member(s) Name	Designation	Signature
1	Mr. Usman Khan	Head, Final Year Projects	
2	Dr. Maaz Bin Ahmed/ Dr. Sohail Imran/ Dr. Muhammad Fahad/ Dr. Ayub Latif	Head of Dept.	
3	Prof Dr Muhammad Khalid Khan	Dean CIS	

Date: 00/00/0000